

Dirac Delta

The formal definition differs from the practical definition of $\delta(t)$, so I won't write them :)

Here are some useful theorems with $\delta(t)$

Theorem 1 (Relationship with $u(t)$)

1.
$$u(t - \alpha)' = \delta(t - \alpha)$$

2.
$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

Theorem 2 (Multiplication)

1.
$$f(t)\delta(t) = f(0)\delta(t)$$

2.
$$f(t - \beta)\delta(t - \alpha) = f(\alpha - \beta)\delta(t - \alpha)$$

Theorem 3 (Convolution)

1.
$$\delta(t - \alpha) * \delta(t - \beta) = \delta(t - \alpha - \beta)$$

2.
$$f(t - \beta) * \delta(t - \alpha) = \int f(\alpha - \beta)\delta(t - \alpha) d\tau = f(t - \alpha - \beta)$$

Remark 4 (Time shifts are swappable in a convolution:)

$$x(t + \alpha) * y(t + \beta) = \int x(\tau + \alpha)y(t + \beta - \tau) d\tau = \int x(\tau)y(t + \beta + \alpha - \tau) d\tau$$

Convolution

To extend and generalise [Theorem 3](#) and keeping in mind [Remark 4](#)

Theorem 5

Writing

$$x(t + \alpha) * y(t + \beta) = x(t + \alpha + \beta) * y(t)x(t) * y(t + \alpha + \beta)$$

is completely equivalent to

$$\int x(\tau + \alpha)y(t + \beta - \tau)d\tau = \int x(\tau + \alpha + \beta)y(t - \tau)d\tau = \int x(\tau)y(t + \alpha + \beta - \tau)d\tau$$

Periodic Signals

Theorem 6 (Convolutional representation of a periodic signal)

$$\text{rep}_T(x(t)) = \sum_{-\infty}^{\infty} x(t - kT) = \sum_{-\infty}^{\infty} \delta(t - kT)x(t)$$

Theorem 7 (Energy and Power of periodic signals)

Due to the nature of a periodic signal its energy is always equal to 0 since it doesn't attenuate

The power is equal to the mean energy over one period

$$P(x(t)) = \frac{\int_{[T]} |x(t)|^2}{T}$$

Remark 8

This is valid also for any function given in a form of $x(t) = x_p(t) + x_{lp}(t)$ with $x_p(t)$ a periodic function and $x_{lp}(t)$ a limited Energy function, then the power will be

$$P(x(t)) = \frac{\int_{[T]} |x_p(t)|^2}{T}$$

since $P(x_{lp}(t)) = 0$

LTI Systems

Definition 9 (Impulse Response)

Given an LTI system, the impulse response of such system is

$$h(t) := L(\delta(t))$$

Fourier Series

Theorem 10 (Transformations of a_k)

Let $x(t) \in L^1$ admit Fourier series a_k

Transformations		Remarks
Conjugate	$\overline{x(t)} \rightarrow \overline{a_{-k}}$	if $x(t) \in \mathbb{R} \implies x(t) = \overline{x(t)} \implies a_k = \overline{a_{-k}}$
Time inversion	$x(-t) \rightarrow a_{-k}$	if $x(t)$ even $\implies x(t) = x(-t) \implies a_k = a_{-k}$
Time scale	$x(\alpha t) \rightarrow a_k \alpha > 0$	$T' = T/\alpha, w' = \alpha w$
Traslation	$x(t + \beta) \rightarrow e^{jk w_0 \beta} a_k$	
Modulation	$e^{jM w_0 t} \rightarrow a_{k-M}$	
Derivation	$\frac{dx(t)}{dt} \rightarrow jk w_0 a_k$	

Theorem 11 (Coefficients for LTI systems)

Let $x(t)$ admit coefficients a_k and let $h(t) \in L^1(\mathbb{R})$, then $y(t) = h * x(t)$ will have the following coefficients:

$$b_k = H(jw_k) a_k$$

Proof.

$$\text{Simply } x(t) = \sum_{-\infty}^{\infty} a_k e^{jw_k t} \text{ then } H(jw) x(t) = \sum_{-\infty}^{\infty} H(jw) a_k e^{jw_k t}$$

□

Theorem 12 (Coefficients of $x(t) = \text{rep}_T(\text{rect}(t/t_0))$)

Let $x(t) = \text{rep}_T(\text{rect}(t/t_0))$ with $t_0 \leq T$, then the coefficients a_k are:

$$a_k = \frac{t_0}{T} \text{sinc}\left(\frac{w k t_0}{2\pi}\right)$$

Proof.

By definition

$$a_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) e^{-jw_k t} dt$$

Let's compute only the integral for now, and since $t_0 \leq T$

$$\int_{-\frac{t_0}{2}}^{\frac{t_0}{2}} e^{-jw_k t} dt = \frac{-1}{jw_k} (e^{-jw_k t}) \Big|_{-\frac{t_0}{2}}^{\frac{t_0}{2}} = \frac{-1}{jw_k} (e^{-jw_k \frac{t_0}{2}} - e^{jw_k \frac{t_0}{2}}) = \frac{1}{jw_k} (-e^{-jw_k \frac{t_0}{2}} + e^{jw_k \frac{t_0}{2}})$$

by using Euler's equation

$$\frac{2}{wk} \sin(wk \frac{t_0}{2}) = \frac{2}{wk} \sin(\pi \frac{wkt_0}{2\pi}) = \frac{2}{wk} \sin(\pi \frac{wkt_0}{2\pi}) = t_0 \text{sinc}(\frac{wkt_0}{2\pi})$$

where the last step was possible by multiplying by $\frac{2t_0\pi}{2t_0\pi}$

□

Theorem 13 (Parseval)

$$\text{Let } x(t) = \sum a_k e^{jk\omega_0 t} \implies E(x(t)) = T \sum |a_k|^2$$

More in particular, if $y(t) = \sum b_k e^{jk\omega_0 t}$ has same period as $x(t)$, then

$$E(x(t), y(t)) = T \sum a_k \overline{b_k}$$

Fourier Transform

Discrete (DFT)

Let's start with the discrete case

Continuous (FT)

Definition 14 (Fourier transform)

Let $x(t)$ be a signal with $t \in \mathbb{R}$, then the Fourier transform is defined as

$$X(w) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Theorem 15 (Existence of $X(w)$)

$$\text{Let } x(t) \in L^1(\mathbb{R}) \implies \exists X(w)$$

But clearly if $x(t)$ is periodic, then $x(t) \notin L^1(\mathbb{R})$.

For $x(t) = e^{j\omega_0 t} \implies X(w) = 2\pi\delta(w - \omega_0)$ and by Fourier series every periodic signal can be written as $x(t) = \sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t}$ with $\omega_k = k\omega_0$, then clearly $X(w) = 2\pi \sum_{-\infty}^{\infty} a_k \delta(w - \omega_k)$

Theorem 16 ($A(x(t))$ using Fourier Transform)

Let $\mathcal{F}(x(t)) = X(w)$, then

$$A(x(t)) = X(w) \big|_{w=0}$$

Proof.

By definition and by setting $w = 0$, $A(x(t)) = \int_{-\infty}^{\infty} x(t) dt = X(0) = \int_{-\infty}^{\infty} x(t) e^{-j0t} dt$

□

With the same logic it is possible to extend the latter theorem to $E(x(t))$.

Theorem 17 (Parseval Theorem)

Let $x(t) \in L^2$ and $\mathcal{F}(x(t)) = X(w)$, then

$$E(x(t)) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(w)|^2 dw$$

More in general, if $y(t) \in L^2$ and $\mathcal{F}(y(t)) = Y(w)$, then

$$E(x(t), y(t)) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(w) \overline{Y(w)} dw$$

Sampling

Sampling consists in the study of a sequence of data $x(kT)_{k \in \mathbb{Z}}$ and the reconstruction of the original signal $x(t)$

Definition 18 (Bandwidth)

The bandwidth of a signal $x : \mathbb{R} \rightarrow \mathbb{R}$ is the length of the positive real part of its fourier transform $B_x := \{w \geq 0 \mid X(jw) > 0\}$

In particular if $B_x \subset [0, w_M]$ with $w_M := \sup_w(B_x)$ then it has limited bandwidth

Example: $\mathcal{F}(\text{sinc}(t)) = \text{rect}(\frac{w}{2\pi})$ which has $w_M = \pi$ has limited bandwidth

Example: $\mathcal{F}(\text{rect}(t)) = \text{sinc}(\frac{w}{2\pi})$ has $w_M = +\infty$

A simple way to write $x_p(t)$ is $x_p(t) = x(t)p(t) = x(t) \sum \delta(t - kT) = \sum x(kT) \delta(t - kT)$

Theorem 19 (Shannon)

Let $x(t)$ be a limited bandwidth signal, then if its sampling pulsation $w_s > 2w_M$, $x_p(t)$ determines $x(t)$

Proof.

1. if $x(t) \in L^1(\mathbb{R})$ its transform exists and is limited
2. if $w_s \geq 2w_M$ then $X_p(w) = \frac{1}{T} \text{rep}_{w_s}(X(w))$ is made of infinite copies
3. filter $x_p(t)$ with a low-pass filter $H_{lp} = T \text{rect}(\frac{w}{2w_c})$ with $w_M < w_c < w_s - w_M$

4. the resulting signal is $X(w)$

□

Theorem 20 (Shannon Interpolator)

By Shannon it is possible to reconstruct $x(t)$ with the following formula

$$x(t) = \sum_{k=-\infty}^{\infty} x(kT) \text{sinc}\left(\frac{t-kT}{T}\right)$$

the series converges in L^2

anche perchè $k \in \mathbb{N}$ quindi se $\alpha \in \mathbb{R}$ allora $\exists k = \alpha \notin \mathbb{N}$ ma se w è comune a tutti gli esponenti allora $k = \frac{n\alpha}{w} \in \mathbb{N}$ dalle proprietà dei segnali periodici

Laplace Transform

Theorem 21 (Laplace transform of $\frac{d}{dt}x(t)$)

This theorem is valid only for unilateral Laplace Transforms.

$$\mathcal{L}\left(\frac{d}{dt}x(t)\right) = sX(s) - x(0-)$$

In general, the n -th derivative has the following transform

$$\mathcal{L}\left(\frac{d^n}{dt^n}x(t)\right) = s^n X(s) - s^{n-1}x(0-) - s^{n-2}x'(0-) - \dots - x^{(n-1)}(0-)$$

Theorem 22 (BIBO stability of $H(s)$)

Let a LTI system admit $H_+(s)$ that admits decomposition of form $\frac{b(s)}{a(s)}$ with a, b polynomials of degree $m \leq n$, then the system is BIBO stable if the roots of $a(s)$ have all $\text{Re}(a) < 0$